SmartMine AI

From simulated mine sensors to actionable safety awareness.

An explainable, containerized proof of concept built on OpenEgiz.

MONITOR • DETECT • VISUALIZE

Copper mine | MQTT | Eclipse Ditto | InfluxDB | Grafana

The problem

01 **Fragmented measurements**

Temperature, methane and ventilation signals need a unified view.

02 **Slow situational awareness**

Risk indicators should be visible as measurements change.

03 **Complex infrastructure**

A reproducible deployment lowers the barrier to testing ideas.

Our solution

One simulated underground copper mine. One live digital twin.

ENVIRONMENT

Temperature / methane / humidity

VENTILATION

Fan status / airflow

SAFETY

LOW / MEDIUM / HIGH

OPERATIONS

Workers underground

Synthetic data only; no connection to real mine equipment.

How the system works

Python
Simulator

→

MQTT
Mosquitto

→

OpenEgiz
+ Eclipse Ditto

→

InfluxDB
Time-series

→

Grafana
Dashboard

Sensor readings are published, processed, stored and visualized.

Docker Compose provides a repeatable local deployment.

Explainable risk detection

HIGH	Methane ≥ 1.0% OR temperature ≥ 30°C
MEDIUM	Methane ≥ 0.5% OR temperature ≥ 27°C
LOW	Otherwise

Illustrative thresholds, not operational safety limits.

Live monitoring & response

Grafana dashboard: SmartMine AI | Underground Mine Safety

METHANE

Concentration (%)

TEMPERATURE

Degrees Celsius

RISK LEVEL

LOW / MEDIUM / HIGH

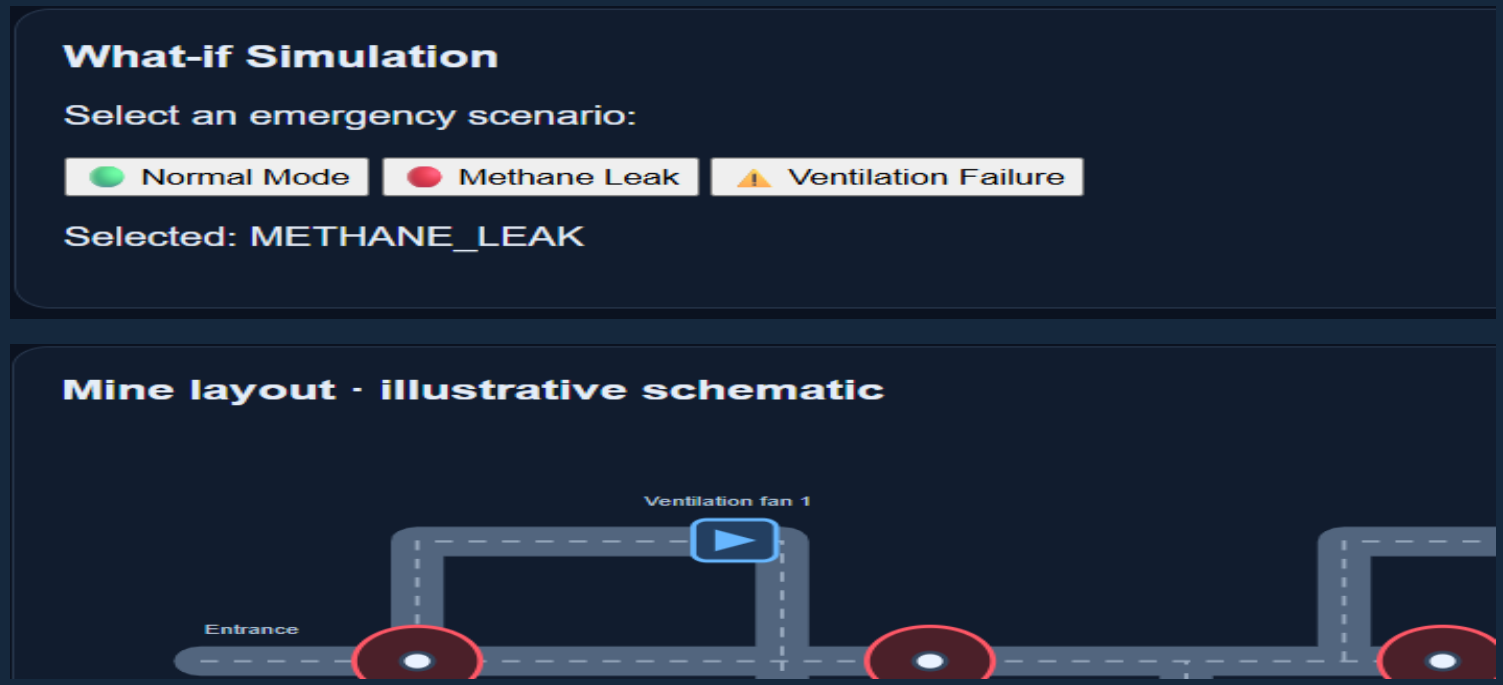
AIRFLOW

Ventilation (m^3/s)

When simulated conditions worsen, the ventilation model increases airflow.

Interactive underground mine map

Live 2D schematic with MQTT telemetry and risk-based zone colors.



WHAT YOU SEE

Temperature / methane

Airflow / safety risk

Ventilation fan status

SCOPE

3 illustrative zones

Shared mine-wide risk

What-if emergency simulation

Choose a scenario on the map; the simulator changes its actual MQTT output.

NORMAL MODE Automatic synthetic readings	Fan ON or BOOST; variable risk
METHANE LEAK Methane rises to 1.5–1.8%	HIGH risk; ventilation BOOST
VENTILATION FAILURE Fan switches OFF; airflow = 0	HIGH risk; emergency condition

Browser button → Map API → MQTT command → Python simulator → Live dashboards

Ready to reproduce

One-command project startup

```
bash team/smartmine/start.sh
```

Starts OpenEgiz, initializes the Ditto twin, installs the Grafana dashboard and launches the simulated telemetry stream.

Repository: github.com/openegiz-contest/team-092

Prototype scope: synthetic telemetry + rule-based detection.